First numerical demonstration of negative density-weighted virial work in three ultra-faint dwarfs under an unregistered Hessian-augmented field law

Won Shik Paik

Independent Researcher, Auckland, New Zealand

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Abstract

This short technical note records the first numerical demonstration, within the UDCT series, of strongly negative density-weighted virial work in three Milky Way ultra-faint dwarfs (Leo IV, Pegasus III and Hydrus I). The result applies to one unregistered, smooth, Hessian-augmented field law whose interpolating multiplier ν and extra tensor P_{ij} are derived from a single action integral. The negative sign is not produced by a negative interpolating function; it arises because the action-derived Hessian contribution exceeds the positive leading term. Standard MOND-family theories (AQUAL, QUMOND) do not contain this extra Hessian-sourced tensor. The observation is therefore a diagnostic for this specific candidate, not a general statement about modified gravity or standard MOND. The calculations were previously reported in UDCT v7.06 and are accompanied here by reproducibility code.

1. Scope and limits

The object under examination is not standard AQUAL or QUMOND. It is one unregistered smooth field law in which a tidal invariant constructed from the Newtonian Hessian sources an additional tensor P_{ij}, with both the multiplier ν and the tensor obtained from the same action. The calculations use a local second-order Taylor environment of the Galactic field (visible baryons only) and idealized stellar profiles (Plummer and a tapered cusp). They do not constitute a self-consistent three-dimensional equilibrium solution with an observed stellar distribution function.

A negative toy-model integral is an obstruction inside that model. It is not a measured significance level against observations and not a theorem about every modified-gravity completion.

2. Numerical result

At the original transition scale, the normalized density-weighted virial work V/V_N is negative for all three objects and both density profiles, and for switch exponents $p = 1–8$. Representative baseline values (Plummer / tapered cusp) are:

- Leo IV: –683.84 / –969.06
- Pegasus III: –810.90 / –1489.62
- Hydrus I: –415.42 / –299.67

Joint scaling of the external acceleration and Hessian shows that the global work changes sign only when both are reduced to roughly 0.3 %–10.6 % of their baseline strength. Raising

the transition scale by a large factor restores positive work, but the field law then approaches the QUMOND form. Even on the positive-work branch, a conditional spherical-isotropic Jeans calculation yields projected dispersions well below the archived values.

3. Relation to prior literature

No earlier published calculation was found that reports a large negative density-weighted virial work arising from an action-consistent Hessian-sourced tensor of this type in ultra-faint dwarfs (or comparable low-acceleration systems). Standard MOND virial relations are positive. Screening/antiscreening studies of MOND fields address different questions. The numerical demonstration reported here (and previously consolidated in UDCT v7.06) is therefore the first of its kind for this specific unregistered candidate.

4. Reproducibility and links

The accompanying Python script reproduces the local-field calculations. It is not an observational detection, not a rerun of any historical preregistered proxy, and not an evaluation of AQUAL or QUMOND.

This note: GitHub https://github.com/wspaik5/First_Negative_Virial_Work_UFDs · Zenodo https://doi.org/10.5281/zenodo.23410285
Original consolidated record: UDCT v7.06, Zenodo https://doi.org/10.5281/zenodo.23039754 · GitHub https://github.com/wspaik5/UDCT_v7.06_Hessian_Term_Sign_Audit

5. Selected references (post-registration emphasis, v7.06–v7.12, plus v7.13 pre-registration)

Paik, W. S. (2026). UDCT v7.06: A sign audit of a Hessian term absent from the standard MOND equation. Zenodo. https://doi.org/10.5281/zenodo.23039754

Paik, W. S. (2026). UDCT v7.07: Structural tests of reciprocal projection, three-dimensional actions, and solar-system limits. Zenodo. https://doi.org/10.5281/zenodo.23130367

Paik, W. S. (2026). UDCT v7.08: Prior-art correction on the action-derived Hessian source term. Zenodo. https://doi.org/10.5281/zenodo.23117541

Paik, W. S. (2026). UDCT v7.09 Post-registration: Positive global virial work with local Jeans obstructions. Zenodo. https://doi.org/10.5281/zenodo.23130843

Paik, W. S. (2026). UDCT v7.10 Post-registration: Global virial sign reversal and local Jeans obstructions. Zenodo. https://doi.org/10.5281/zenodo.23233837

Paik, W. S. (2026). UDCT v7.11 Post-registration: Constant-anisotropy Jeans and outer-boundary sensitivity in Hydrus I. Zenodo. https://doi.org/10.5281/zenodo.23263099

Paik, W. S. (2026). UDCT v7.12 Post-registration. Zenodo. https://doi.org/10.5281/zenodo.23296444

Paik, W. S. (2026). UDCT v7.13 Preregistration: Refined Jeans sign-boundary and Hessian diagnostics in Hydrus I. Zenodo. https://doi.org/10.5281/zenodo.23342971

This note does not register a new constitutive law. It isolates and records one clear numerical outcome of the earlier audit series under explicitly limited assumptions.